PARENT NOTIFICATION

URL:
https://docs.google.com/spreadsheets/d/12mmfsrlmgqYG35CU8tzj5jwjoTbO9gjrJjlvvnhJPH4/edit?gid=969653990#gid=969653990

CODE:

```
#include <WiFi.h>

#include <WiFiClientSecure.h>

#include <HTTPClient.h>

#include <Adafruit_Fingerprint.h>

#include <ArduinoJson.h>


// -------- WiFi Credentials --------

const char* WIFI_NAME = "Starlit";

const char* WIFI_PASSWORD = "password";


// -------- Google Apps Script URL --------

const char* SCRIPT_URL =
"https://script.google.com/macros/s/AKfycbyTHaK91ubMCGJw_8umR3mp51ZmHo8Le4Lpd08Xtnk2dmotQg23TIvLoxDVsKI2SyyN/exec";


// -------- Fingerprint Sensor --------

// ARRIVED for IN ESP32

// LEFT for OUT ESP32

const char* EVENT_TYPE = "ARRIVED";


HardwareSerial FingerSerial(2);

Adafruit_Fingerprint finger = Adafruit_Fingerprint(&FingerSerial);


unsigned long lastScanTime = 0;

const unsigned long SCAN_DELAY = 3000;
```

```
int lastFingerprintID = -1;
unsigned long lastFingerprintTime = 0;
const unsigned long DUPLICATE_DELAY = 5000;

void setup() {
  Serial.begin(115200);
  delay(1000);

  Serial.println();
  Serial.println("HostelMate Fingerprint Attendance");
  Serial.println("--------------------------------");

  // R307S TX -> ESP32 GPIO16 (RX2)
  // R307S RX -> ESP32 GPIO17 (TX2)
  FingerSerial.begin(57600, SERIAL_8N1, 16, 17);
  finger.begin(57600);

  if (finger.verifyPassword()) {
    Serial.println("Fingerprint sensor connected!");
  } else {
    Serial.println("Fingerprint sensor not detected!");
    Serial.println("Check TX, RX, power and baud rate.");

    while (true) {
      delay(1000);
    }
  }
```

```
  connectWiFi();
}

void loop() {
  if (WiFi.status() != WL_CONNECTED) {
    connectWiFi();
  }

  if (millis() - lastScanTime >= SCAN_DELAY) {
    lastScanTime = millis();

    int fingerprintID = getFingerprintID();

    if (fingerprintID > 0) {
      // Prevent repeated logging of the same finger
      // during a short interval.
      if (fingerprintID == lastFingerprintID &&
          millis() - lastFingerprintTime < DUPLICATE_DELAY) {
        return;
      }

      lastFingerprintID = fingerprintID;
      lastFingerprintTime = millis();

      Serial.print("Fingerprint ID: ");
      Serial.println(fingerprintID);

      sendToGoogleSheets(fingerprintID);
    }
```

```
  }
}

void connectWiFi() {
  if (WiFi.status() == WL_CONNECTED) {
    return;
  }

  Serial.print("Connecting to WiFi...");

  WiFi.mode(WIFI_STA);
  WiFi.begin(WIFI_NAME, WIFI_PASSWORD);

  int attempts = 0;

  while (WiFi.status() != WL_CONNECTED && attempts < 30) {
    delay(500);
    Serial.print(".");
    attempts++;
  }

  if (WiFi.status() == WL_CONNECTED) {
    Serial.println("\nWiFi connected!");
    Serial.print("IP address: ");
    Serial.println(WiFi.localIP());
  } else {
    Serial.println("\nWiFi connection failed!");
  }
}
```

```
int getFingerprintID() {
  uint8_t result = finger.getImage();

  if (result != FINGERPRINT_OK) {
    return -1;
  }

  result = finger.image2Tz();

  if (result != FINGERPRINT_OK) {
    Serial.println("Fingerprint image conversion failed.");
    return -1;
  }

  result = finger.fingerSearch();

  if (result == FINGERPRINT_OK) {
    Serial.println("Fingerprint matched!");
    return finger.fingerID;
  }

  if (result == FINGERPRINT_NOTFOUND) {
    Serial.println("Fingerprint not enrolled.");
  } else {
    Serial.println("Fingerprint matching error.");
  }

  return -1;
```

```
}

void sendToGoogleSheets(int fingerprintID) {
  if (WiFi.status() != WL_CONNECTED) {
    Serial.println("WiFi disconnected. Data not sent.");
    return;
  }

  WiFiClientSecure client;

  // Testing only. Use certificate validation for production.
  client.setInsecure();

  HTTPClient https;
  https.setFollowRedirects(HTTPC_STRICT_FOLLOW_REDIRECTS);
  https.setTimeout(15000);

  if (!https.begin(client, SCRIPT_URL)) {
    Serial.println("Could not open Apps Script URL.");
    return;
  }

  https.addHeader("Content-Type", "application/json");

  StaticJsonDocument<128> request;

  request["fingerprintId"] = fingerprintID;
  request["eventType"] = EVENT_TYPE;
```

```
String requestBody;
serializeJson(request, requestBody);

Serial.println("Sending attendance to Google Sheets...");
Serial.println(requestBody);

int httpCode = https.POST(requestBody);

if (httpCode > 0) {
  String response = https.getString();

  Serial.print("HTTP response code: ");
  Serial.println(httpCode);

  Serial.println("Server response:");
  Serial.println(response);

  StaticJsonDocument<2048> result;
  DeserializationError error = deserializeJson(result, response);

  if (!error && result["success"] == true) {
    Serial.println("Attendance saved successfully!");

    Serial.print("Fingerprint ID: ");
    Serial.println(fingerprintID);

    Serial.print("Student Name: ");
    Serial.println(result["studentName"] | "Unknown");
  } else {
```

```
      Serial.println("Google Apps Script returned an error.");
    }
  } else {
    Serial.print("HTTP request failed: ");
    Serial.println(https.errorToString(httpCode));
  }

  https.end();
}
```